LESSON 01 / 02

Status of this sheet

The equation $x^y = y^x$ and multi-base logarithmic systems are **standard IOQM/RMO algebra** and appear in specialised exam questions; everything below is derived from the Part I toolkit (monotonicity, change of base, $\frac{\ln t}{t}$) with no new theory assumed. Nothing here "can never appear in JEE" — the classification is only that these models go beyond the current syllabus sentence "Logarithms and their properties".

$x^y = y^x$ — the complete positive solution set

For $x, y > 0$: if $x = y$ everything works. If $x \neq y$, take ln: $y\ln x = x\ln y$, i.e. $\dfrac{\ln x}{x} = \dfrac{\ln y}{y}$.

So x, y are two distinct points where $f(t) = \frac{\ln t}{t}$ takes the **same value** — and f increases on $(1, e]$, decreases on $[e, \infty)$, so any level is cut at most twice.

SHORT DERIVATION

Parametrisation of all unequal solutions

Put $y = tx$ with $t > 0$, $t \neq 1$. Then $x^{tx} = (tx)^x \Rightarrow x^t = tx \Rightarrow x^{t-1} = t \Rightarrow x = t^{1/(t-1)}$, $y = t^{t/(t-1)}$. Writing $t = 1 + \dfrac{1}{n}$ gives the classic family $x = \left(1 + \dfrac{1}{n}\right)^n$, $y = \left(1 + \dfrac{1}{n}\right)^{n+1}$ for $n > 0$ ($n \in \mathbb{Q}^+$ or $\mathbb{N}$). Check $n = 1$: $(x, y) = (2, 4)$: $2^4 = 4^2 = 16$ ✓. Integer solutions with $x \neq y$: exactly $(2, 4)$ and $(4, 2)$.

WORKED EXAMPLE

Worked. Real solutions with $x < y$, $xy = 8$, $x^y = y^x$?

$\frac{\ln x}{x} = \frac{\ln y}{y}$ with $y = 8/x$: solve $\frac{\ln x}{x} = \frac{\ln(8/x)}{8/x}$ numerically/graphically by monotonicity arguments; or use the parametrisation: $\frac{y}{x} = t = 1 + \frac{1}{n}$, $xy = 8$ fixes n. From $y = tx$ and $xy = 8$: $tx^2 = 8$. With $x = t^{1/(t-1)}$ this becomes $t^{1 + \frac{2}{t-1}} = 8 \Rightarrow t^{\frac{t+1}{t-1}} = 8$. Trial $t = 4$: $4^{5/3} \approx 10.08 \neq 8$; $t = 2$: $2^3 = 8$ ✓. So $t = 2$, $x = 2$, $y = 4$. Answer: $(x, y) = (2, 4)$ (and the symmetric pair if order is free).

Multi-base logarithmic systems

The three moves: (1) reciprocity $\log_a b \cdot \log_b a = 1$; (2) chain $\log_a b \cdot \log_b c = \log_a c$; (3) change every base to ln (or to a common base c) and clear denominators.

WORKED EXAMPLE

Worked. Solve $\log_2 x = \log_x 4$

Domain $x > 0$, $x \neq 1$. *Reciprocity:* $\log_x 4 = \dfrac{\ln 4}{\ln x} = \dfrac{2\ln 2}{\ln x}$; *also* $\log_2 x = \dfrac{\ln x}{\ln 2}$. *Set* $u = \ln x$: $\dfrac{u}{\ln 2} = \dfrac{2\ln 2}{u} \Rightarrow u^2 = 2(\ln 2)^2 \Rightarrow u = \pm \ln 2 \cdot \sqrt{2}$. *So* $x = 2^{\sqrt{2}}$ *or* $x = 2^{-\sqrt{2}}$. *Check both are* > 0 *and* $\neq 1$ ✓. *Answer:* $x \in \{2^{\sqrt{2}},\ 2^{-\sqrt{2}}\}$.

Towers obey a convention: $a^{b^c} = a^{(b^c)}$ — *exponentiation is* ***right-associative***; $(a^b)^c = a^{bc} \neq a^{b^c}$ *in general. Nested logs peel from the inside:* $\log_2(\log_3(\log_4 256)) = \log_2(\log_3 4) = \log_2\left(\dfrac{\ln 4}{\ln 3}\right)$ — *leave as exact expressions; never invent a "log of log" law.*

COMMON TRAP

Traps on this sheet

- $x^y = y^x$ *over* ***integers*** *vs* ***reals*** *give different answer sets; the parametrisation covers reals; integers need* $t = 1 + 1/n$ *with* t *rational* $\Rightarrow n = 1$ *only among integers (for* $x < y$ *positive integers:* $\{2, 4\}$*).*
- $\log_a b = \log_a c \Rightarrow b = c$ ***needs the same base***; $\log_2 x = \log_3 x \Rightarrow x = 1$ *only.*
- *Solving* $\log_x(\cdot)$ *adds base conditions* $x > 0$, $x \neq 1$ *on top of argument positivity.*
- $f(t) = \frac{\ln t}{t}$ *is* ***not*** *monotone on* $(0, \infty)$ — *it turns at* $t = e$*; two solutions for* $f(t) = c \in (0, 1/e)$ *sit on opposite sides of* e.

Assumptions & conventions

All variables positive reals unless stated; integer solutions are a separate reading of the same equations; $\frac{\ln t}{t}$ *analysis uses its stated monotonicity shape (derivable later by calculus; here taken as the known shape of the standard curve, with the level-curve picture as the geometric model).*

LESSON 02 / 02

Status of this sheet

Order-of-magnitude estimates and Legendre's formula are ***classic IOQM/RMO number theory*** *built from the Part I toolkit (logs, floors, geometric progressions). The hierarchy statements below are comparisons between standard functions, models derivable from supplied information — not additional theory claimed as a syllabus item.*

The growth hierarchy (state it as ordering, use it as a tool)

For fixed $a>1$, $b>1$, $c>1$, $k\in\mathbb{N}$, *as* $x\to\infty$:

$$\log_c x \ll x \ll x^k \ll a^x \ll b^{b^x}.$$

Use in estimates: *to compare* $n^{\log n}$*-shaped quantities, take* $\log$ *once or twice until everything is polynomial. Example: which is larger for huge* n *—* 100^n *or* n^{100}*? Take* $\ln$*:* $n\ln 100$ *vs* $100\ln n$*; divide by* n*:* $\ln 100\approx 4.605$ *vs* $\frac{100\ln n}{n}\to 0$*. So* 100^n *wins for large* n*; at* $n=100$ *the two are* ***equal****, for* $n>100$ *the exponential* 100^n *is larger, for* $e<n<100$ *the power* n^{100} *is larger (crossover where* $\frac{\ln n}{n}=\frac{\ln 100}{100}=0.046$*).*

KEY RESULT

Digit-counting engine. $d(N)=\lfloor \lg N\rfloor+1$ *for integers* $N\geq 1$*. Sums of digits in base* b*:* $N=\sum d_i b^i$ *with* $0\leq d_i<b$*; the digit-sum* $s_b(N)=\sum d_i$ *satisfies* $N-s_b(N)$ *is divisible by* $b-1$ *(casting out* $b-1$*'s). This pair (*$\lfloor\log\rfloor$*, digit sum) is exactly what Legendre's formula packages.*

Legendre's formula — the exponent of a prime in $n!$

For prime p*:* $v_p(n!)=\displaystyle\sum_{k=1}^{\infty}\left\lfloor\frac{n}{p^k}\right\rfloor=\frac{n-s_p(n)}{p-1}$*, where* $s_p(n)$ *is the digit sum of* n *in base* p *(finitely many non-zero terms: stop when* $p^k>n$*).*

SHORT DERIVATION

Derivation — count multiples, then clean up with digits

Among $1,2,\ldots,n$*, exactly* $\lfloor n/p\rfloor$ *are divisible by* p*; among those,* $\lfloor n/p^2\rfloor$ *contribute a* ***second*** *factor* p*; and so on — each integer* m *contributes* $v_p(m)$ *factors, so the total is* $\displaystyle\sum_{k\geq 1}\lfloor n/p^k\rfloor$ ✓ *(count each* m *once per power of* p *dividing it). For the closed form, write* $n=s_p(n)+(p-1)\displaystyle\sum_{k\geq 1}\lfloor n/p^k\rfloor$ *(base-*p *identity:* $\lfloor n/p^k\rfloor=\displaystyle\sum_{j>k} d_j p^{j-k-1}$*), rearranged to the formula.*

WORKED EXAMPLE

Worked. $v_2(100!)$ and $v_5(100!)$

$v_2(100!) = 50 + 25 + 12 + 6 + 3 + 1 = 97$. *Via digits:* $100 = (1100100)_2$, $s_2 = 3$, $\frac{100-3}{1} = 97$ ✓. $v_5(100!) = 20 + 4 = 24$; *digits:* $100 = (400)_5$, $s_5 = 4$, $\frac{100-4}{4} = 24$ ✓. *Consequence:* $100!$ *ends in exactly* 24 *zeros (the power of* $10 = 2 \cdot 5$ *is the smaller of* v_2, v_5*).*

Series estimates with logs and floors

Truncation errors: $\sum_{k=1}^{n} \frac{1}{k}$ *grows like* $\ln n$ *(harmonic estimate:* $\ln(n+1) \le H_n \le 1 + \ln n$ *— provable by comparing the sum with* $\int \frac{dx}{x}$*, a standard derived model). The telescoping estimate* $\frac{1}{k+1} \le \ln\left(1 + \frac{1}{k}\right) \le \frac{1}{k}$ *(from* $\frac{1}{1+t} \le \ln(1+t)$*-style log bounds at* $t = 1/k$*) is the workhorse behind it.*

WORKED EXAMPLE

Worked. Bound $H_{100} = 1 + \frac{1}{2} + \cdots + \frac{1}{100}$

$\ln 101 \approx 4.615 \le H_{100} \le 1 + \ln 100 \approx 5.605$; *the true value is* ≈ 5.187 ✓ *inside the bracket. Use such brackets to decide sign/size questions without computing sums.*

COMMON TRAP

Traps on this sheet

- $\lfloor n/p \rfloor + \lfloor n/p^2 \rfloor + \cdots$ *is* **not** *a geometric series — do not sum it as one; each term is a floor.*
- *Trailing zeros of* $n!$ *are* $v_5(n!)$ *(5s are rarer than 2s), not* $v_2(n!)$*.*
- log *estimates compare growth rates; a bound valid "for large* n*" must not be quoted for a specific small* n *without checking (e.g.* $n^{100} > 100^n$ *fails for* $n = 2$*).*
- *Digit sums* s_p *depend on the base;* $s_2(100) = 3$ *counts* ***ones in binary****, not decimal digits.*

Assumptions & conventions

p *prime;* $v_p(m)$ *= exact exponent of* p *in* m*; floors are real floors; harmonic estimates quoted as standard derived bounds (checked numerically above); hierarchy statements are eventual comparisons (there is always a crossover point).*